

# AKKSHAYA KUMAR R V

9787504151 • akkshayakumar2410100@ssn.edu.in • linkedin.com/in/rv-akkshaya-kumar • github.com/akki0312-tech

## EDUCATION

### B.Tech. Information Technology

Sri Sivasubramaniya Nadar College of Engineering

Aug 2024 - Present

8.29 CGPA

### Grade 12, Computer Science

May 2024

Narayana CBSE school, Bengaluru

91.6%

### Grade 10

May 2022

Jaigopal Garodia Rashtrorothana Vidya Kendra, Bengaluru

96.6%

## TECHNICAL SKILLS

### Relevant Coursework:

Problem Solving & Python Programming · Programming in C & Data Structures · Introduction to Web Programming · Object-Oriented Programming using Java · Database Technology · Principles of Software Engineering & Practices · Digital Systems & Microprocessors Design · Introduction to Digital Communication · Pattern Recognition & Machine Learning · Data Analytics & Visualization · Advanced Data Structures & Algorithm Analysis · Signal Processing for Machine Learning · Data Communication & Networks · Computer Organization

### Programming Skills & Tools:

**Languages:** Python, C, Java, SQL, JavaScript

**Frameworks/Tools:** Flutter, Bootstrap, Git, Linux

**ML/Data:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

**Databases:** MySQL, Firestore

**Networking:** Wireshark, Cisco Packet Tracer

**Visualization:** Power BI, Tableau

**AI/Cloud:** LangGraph, Amazon Bedrock, OpenSearch

## ACADEMIC PROJECTS

### Namma Ration

Developed a bilingual (English/Tamil) Public Distribution System mobile application for the Government of Tamil Nadu using Flutter, with Firestore as the real-time backend for order and slot management. Implemented end-to-end features including interactive item pre-ordering, time slot booking, QR-based digital token generation, and a map-based fair price shop locator powered by Google Maps API to streamline ration distribution and enhance accessibility.

### SAR AI Narrative Generator

Built an AI system to automate Suspicious Activity Report (SAR) generation in banking using a multi-agent LangGraph workflow (Investigator, Regulatory Oracle, Narrative Composer), reducing report generation time from 5 hours to 30 minutes. Integrated Llama 3.1 via Amazon Bedrock with a PostgreSQL/OpenSearch RAG pipeline to automatically contextualize money laundering patterns and map them to regulatory violations.

### ECG Arrhythmia Detection Pipeline

Built a Python pipeline to detect cardiac arrhythmias from raw ECG signals by extracting morphological and frequency-domain features using Pan-Tompkins, Wavelet Transforms, STFT/FFT, and Shannon Entropy. Fed the extracted features into a custom diagnostic voting matrix for anomaly classification, and built Matplotlib/Seaborn dashboards to visually explain the reasoning behind each arrhythmia flag without relying on a black-box ML model.

## ACHIEVEMENTS

- GDG TechSprint '26** at SSN College of Engineering, Chennai - **Winner** (Among 80 teams)
- Nourishnovate'26** at REC, Chennai - **Runner up** (Among several teams across South India)
- ProCode Cup'26** at SSN College of Engineering, Chennai – **Special Mention**